Saumya Rishikesh

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Education

Manipal University Jaipur **Jaipur, Rajasthan**
Bachelor of Technology in Computer Science *September 2022 – July 2026*

Work Experience

Software Engineering Intern **Jan 2026 – Present**
DealMatrix Services Pvt Ltd *Remote*

- Developed multilingual AI voice agents supporting **10+ Indian languages** for automated tele-calling, reducing per-call operational costs by **60–70%** vs. human callers with 24/7 availability.
- Architected a low-latency streaming pipeline — **Sarvam AI** (STT/TTS), **Gemini**-primary with **Groq** fallback LLM chain, **LiveKit** over WebRTC — achieving **<1s** end-to-end responses, instrumented per call (TTFT/latency metrics in **PostgreSQL + Grafana**).
- Built self-hosted SIP telephony (**LiveKit SIP → Asterisk PBX → Jio trunk**) with **RBI-compliant** monitoring: automated call recording, transcript logging, and WhatsApp follow-ups.
- Designed modular agent architecture with conversation state management, interruption handling, and silence watchdogs for debt-collection workflows, extensible to customer-service and insurance use cases.

Data Science Intern **June 2025 – July 2025**
SP Software (P) Limited *Hyderabad, India*

- Developed automated data extraction pipelines for internal invoice documents using OCR and deep learning-based layout analysis.
- Fine-tuned **TrOCR** and **Donut** models to extract structured fields from scanned invoices; trained custom **YOLO** models for detecting annotated regions using bounding box supervision.
- Prepared labeled datasets from internal archives; built post-processing logic to clean and validate extracted text using schema constraints.

Student Researcher **Dec 2024 – May 2025**
Manipal University Jaipur *Jaipur, India*

- Researched Alzheimer's disease classification using MRI neuroimaging data, training CNN models such as **ResNet-18, VGG-19, ConvNeXt-Tiny**.
- Implemented multi-phase fine-tuning, augmentation, and **Grad-CAM** interpretability; presented at **ICAIIHI 2025** (IEEE co-sponsored); follow-up research in progress.

Publications

S. Rishikesh et al., "Leveraging CNN Architectures for Alzheimer's Disease Classification Using Neuroimaging Data," in *Proc. 2nd Int. Conf. on Artificial Intelligence for Innovations in Healthcare Industries (ICAIIHI)*, IEEE, 2025. [IEEE Xplore]

Projects

Math Routing Agent | *Python, FastAPI, React + Vite, RAG (Qdrant), LangGraph, DSPy, MCP*

- Developed an **agentic math-tutoring system** using **RAG** over curated datasets (**GSM8K, Orca 200k**) for grounded step-by-step reasoning.
- Engineered a **LangGraph**-orchestrated **multi-agent pipeline** (**Professor, Critic, Guardrail, Human-in-the-Loop**) with LLM-summarized conversation memory and bounded iterative refinement loops.
- Built a **Qdrant** retriever tool and a custom **MCP web-search tool server**; compiled the human-feedback refiner with **DSPy (BootstrapFewShot)** from collected corrections.
- Implemented **async human-in-the-loop** over a **FastAPI** backend — the graph suspends on critic rejection and resumes on human feedback — with a **React + Vite** frontend showing visible reasoning traces.

SHL Assessment Recommendation System | *Python, FastAPI, Streamlit, Qdrant, LLMs* | Live Demo

- Built a hybrid retrieval pipeline (dense embeddings + BM25 sparse + server-side rank fusion) in **Qdrant** with two-stage **reranking** (Cohere, ZeRank) to rank SHL assessments from natural-language queries.
- Developed LLM-based intent parsing to extract skills, duration constraints, and behavioral vs. technical requirements, applied as metadata filters during retrieval.
- Improved mean **Recall@10 by 58%** (0.24 → 0.37) on a labeled query set through retrieval tuning and reranking; deployed live (**Streamlit** UI + **FastAPI** backend on Render).

News Sentiment Analysis and TTS | *Python, FastAPI, Streamlit, Docker, Hugging Face Spaces*

- Developed a news sentiment-analysis and TTS system deployed on **Hugging Face Spaces**: weighted title/body **VADER** scoring with spoken Hindi reports via English→Hindi translation using **MBart-50** (50-language model) and **gTTS**.
- Built **KeyBERT** keyword extraction with cross-article comparative analysis and a real-time **Streamlit** sentiment-visualization and audio-playback UI over a **FastAPI** backend.

Achievements

Dean's List: Achieved Dean's List recognition in both **1st** and **2nd** semesters for academic excellence. View Certificate

Technical Skills

Languages: Python, SQL, JavaScript
Frameworks/Tools: PyTorch, FastAPI, LangGraph, LangChain, DSPy, Docker, Git, React, Streamlit, Hugging Face, LiveKit
Databases/Infra: Qdrant, ChromaDB, PostgreSQL, MongoDB, Grafana
Libraries: NumPy, pandas, Scikit-learn, OpenCV, Transformers, Matplotlib, Librosa